

Method Fit Matrix — E.D.F.C.V.

Deliverable 02 · Research Methods and Scientific Integrity in AI and Advanced Technologies

DRIVING QUESTION — RQ2 (EVALUATION), FROM DELIVERABLE 01 Does forming autonomous judgment through this agent outperform restricting behavior, increasing metacognitive reflection and product integrity and reducing AI dependence, relative to unrestricted and restrictive-normative conditions? RQ1 (design) is answered by building the artifact itself; the candidates below are therefore evaluation strategies for RQ2, not rival paradigms.	PHILOSOPHICAL POSITION AND APPROACH Pragmatism — knowledge is warranted by its usefulness in action (Goldkuhl, 2012). Design Science Research as the research approach (Hevner et al., 2004); intended contribution, a nascent design theory (Gregor & Hevner, 2013).
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FEDS evaluation strategy: Human Risk & Effectiveness (Venable et al., 2016) — the dominant design risk is human, not technical. **Formative / artificial** episodes: expert validation of virtue definitions and pilot annotation of the Spanish corpus. **Summative / naturalistic** episode: the three-condition comparison in live course sections. **Unit of assignment: intact section**, which is the source of the clustering and contamination constraints scored below.

Candidate evaluation method	E — Epistemological fit weight 30	D — Data availability weight 20	F — Feasibility weight 15	C — Contribution type weight 25	V — Venue fit weight 10	Weighted 0–100
M1 · Three-condition quasi-experiment Pre/post, allocation by intact section: agent, restrictive-normative, unrestricted. The Deliverable 01 plan as submitted.	3 Naturalistic summative episode fits FEDS, but a mean delta cannot evidence phronesis.	3 Three outcomes need Spanish-validated instruments; product integrity is still underspecified.	4 Three arms need three or more intact sections; I am designer, instructor and evaluator.	3 Evidences effectiveness only; yields no mechanism, therefore no design principles.	3 Fits Computers & Education; DESRIST expects the design-theory argument it omits.	63.0 2ND
M2 · Three-arm cluster-randomised trial Same three conditions, randomised at section level rather than assigned.	3 Strongest causal warrant, but randomisation strips the naturalistic use this strategy targets.	2 Three arms multiply clusters needed; ICC and cross-section talk erode attainable power.	2 Needs institutional authority to randomise sections across faculty; outside my control.	4 Higher-grade causal evidence; strong stand-alone methodological contribution.	5 Best fit for Computers and Education: AI and IJEI; CONSORT cluster reporting.	62.0 3RD
M3 · Explanatory sequential mixed methods M1 as the quantitative strand, followed by case-anchored qualitative analysis of the annotated virtue corpus. SELECTED	5 Only candidate reaching the particular reasoning phronesis requires; matches pragmatism's warrant in action.	5 Reuses the Spanish virtue-annotated corpus already core to the thesis; primary data by design.	3 Exemplar-anchored coding and per-virtue alpha are a real added cost over M1 alone.	5 Delivers effectiveness and the mechanism a nascent design theory needs as design principles.	4 Strong for IJEI and DESRIST; penalised by eight-page conference formats.	92.0 1ST

Elimination rule ($E \leq 2 \rightarrow$ discarded before any aggregate is computed): no candidate eliminated. Lowest epistemological-fit score is 3 (M1 and M2), so the rule changes nothing here; this is recorded as a result, not an omission. **No candidate dominates another across all five criteria:** M1 leads on feasibility, M2 on venue fit, M3 on the remaining three. The set is therefore genuinely competitive rather than constructed to favour the selection — and because M1 and M2 separate by only 1.0 point, the aggregate alone could not adjudicate between them.

WEIGHTING RATIONALE ($\Sigma = 100$) Highest — E, 30. The object is a dispositional ethical construct. A method that cannot reach the student's reasoning measures something other than what RQ2 asks. An epistemological misfit is unrecoverable once data collection closes, unlike a power shortfall or a venue mismatch, both of which can be repaired afterwards. Lowest — V, 10. Venue fit follows from methodological quality and can be re-targeted by partitioning results across papers. Choosing a method to suit a journal inverts means and ends — the same vice this study attributes to students who delegate the task rather than the judgment. Intermediate: C 25 (DSR requires prescriptive knowledge, not only a working artifact), D 20, F 15.	WHY NOT THE ALTERNATIVES Not M2 (cluster RCT). Rejected on availability and feasibility, not on merit: it outscores the selected design on venue fit, 5 against 4, and is competitive on contribution. Its causal advantage is unreachable with the sections I can obtain, and is partly cancelled anyway, because randomising allocation does not stop students talking across sections. Not M1 (quantitative only). It establishes effect without mechanism, so it cannot yield the design principles that make this a design theory rather than a programme evaluation. M1 is the more feasible option taken alone (F 4 against 3), and that difference is the honest price of the mechanism. M1 is not discarded: it is retained in full as the quantitative strand nested inside M3.	OPEN METHODOLOGICAL TENSIONS Positionality. I am designer, instructor and evaluator of my own sections. Consent must be decoupled from grading and log custody delegated to a third party; ethics approval is a precondition, not a formality. Contamination. Students talk across sections, so the restrictive-normative condition can leak. A manipulation check is required; randomisation would not remove this. Reliability–validity (unresolved). Operationalising prudence raises agreement but drifts toward the rulebook this study contests. Mitigated by exemplar-anchored rather than rule-based annotation, with Krippendorff's alpha reported per virtue instead of pooled (Hayes & Krippendorff, 2007). Whether exemplar anchoring preserves the construct without importing the same rule logic is still open.
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DECISION I will use <u>an explanatory sequential mixed-methods evaluation</u> — the three-condition quasi-experiment followed by case-anchored qualitative analysis of the annotated virtue corpus — because it is the only candidate that both tests effectiveness and yields the design principles a nascent design theory requires, and I reject <u>the three-arm cluster-randomised trial</u> because its causal advantage is unattainable with the sections I control and is partly cancelled by cross-section contamination.
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References. Goldkuhl, G. (2012). Pragmatism vs interpretivism in qualitative information systems research. *European Journal of Information Systems*, 21(2), 135–146. <https://doi.org/10.1057/ejis.2011.54> · Gregor, S., & Hevner, A. R. (2013). Positioning and presenting design science research for maximum impact. *MIS Quarterly*, 37(2), 337–355. <https://doi.org/10.25300/MISQ/2013/37.2.01> · Hayes, A. F., & Krippendorff, K. (2007). Answering the call for a standard reliability measure for coding data. *Communication Methods and Measures*, 1(1), 77–89. <https://doi.org/10.1080/19312450709336664> · Hevner, A. R., March, S. T., Park, J., & Ram, S. (2004). Design science in information systems research. *MIS Quarterly*, 28(1), 75–105. <https://doi.org/10.2307/25148625> · Venable, J., Pries-Heje, J., & Baskerville, R. (2016). FEDS: a framework for evaluation in design science research. *European Journal of Information Systems*, 25(1), 77–89. <https://doi.org/10.1057/ejis.2014.36>